

2	(a)	(i)	$\Delta p = \Delta(mv) = m \Delta v = m (v_f - v_i)$ $= 1.2 (-0.8 - 4)$ $= -5.76 \text{ kg m s}^{-1}$	
		(ii)	Ball S must have a change of momentum of 5.76 kg m s^{-1} as well, thus $5.76 = m (v_f - v_i)$ $= 3.6 (v_f - 0)$ $v_f = 1.6 \text{ m s}^{-1}$	
	(b)		Relative speed of approach $= u_1 - u_2 = 4 - 0 = 4 \text{ m s}^{-1}$ relative speed of separation $= v_2 - v_1 = 1.6 - (-0.8) = 2.4 \text{ m s}^{-1}$ Since they are different, this is an inelastic collision. (accept ans using KE conservation. $KE_i = 9.60 \text{ J}$, $KE_f = 4.99 \text{ J}$)	